TetraMesh: Linked Tetrahedra as Universal 3D Primitives for Multimodal DNA Computing

DOGMA Paper Series — Paper III

Wenyan Qin*
MapleAI Intelligence Inc.
Toronto, ON, Canada

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Abstract

Triangle meshes have dominated 3D representation for four decades, yet they are fundamentally incomplete as computational primitives: they lack volume, chirality, and complete graph connectivity. We introduce **TetraMesh**—a linked-tetrahedron representation where each tetrahedron maps to a DNA base quartet (A, C, G, T), each of its 6 edges forms a K_4 complete graph encoding pairwise hydrogen-bond affinities, and each cell carries signed volume and chirality matching B-form DNA. We prove that TetraMesh provides **4× information density**, **topological completeness** (K_4 vs. K_3), and **chirality preservation** that no triangle-based representation achieves. Adjacent tetrahedra share triangular faces, forming **LinkedTetraChains** mirroring the sugar-phosphate backbone; two complementary chains form a **DoubleHelixMesh** with Watson-Crick cross-links. We present differentiable pipelines converting images, video, audio, and triangle meshes into TetraMesh tokens consumable by the DOGMA genomic computation architecture [1], establishing a **universal multimodal encoding** where all modalities share the same DNA-native representation. We prove a **Universal Approximation Theorem for tetrahedral encodings** and demonstrate that the K_4-aware DifferentiableTetraOps with volume gating achieve formally superior expressiveness over K_3-based alternatives. TetraMesh is the first 3D primitive designed from molecular biology principles, bridging four decades of computer graphics with the emerging paradigm of genomic computation.

*Corresponding author. Email: wenyan@mapleai.org

1 Introduction

The triangle mesh is the standard 3D primitive in computer graphics, computer vision, and geometric deep learning [3, 4]. Yet triangles have fundamental limitations as computational primitives:

1. **No volume.** A triangle is a 2D surface element. Volume must be reconstructed indirectly via watertight surfaces.
2. **No chirality.** A triangle has no intrinsic handedness; orientation is conventionally imposed via vertex winding order.
3. **No biological mapping.** There is no natural correspondence between 3 vertices and the 4-letter DNA alphabet $\{A, C, G, T\}$.
4. **Incomplete connectivity.** Three vertices form K_3 with only 3 edges. The K_4 complete graph requires 4 vertices and 6 edges—capturing all pairwise interactions.

We introduce **TetraMesh**: a 3D representation built from linked tetrahedra that resolves all four limitations simultaneously. Each tetrahedron provides:

- **4 vertices** ↔ DNA bases (A, C, G, T)
- **6 edges** ↔ K_4 complete graph (all pairwise interactions)
- **4 triangular faces** ↔ shared information surfaces
- **Signed volume** ↔ scale and chirality ($\chi = \pm 1$)

This creates a natural bridge between 3D geometry and DNA computing: any 3D structure can be encoded as a sequence of DNA base quartets with biologically meaningful edge weights. The key insight is that the tetrahedron is the *minimal simplex with volume*—just as the DNA base quartet (A, C, G, T) is the minimal alphabet of life.

2 Formal Foundations

We establish the mathematical foundations of TetraMesh through a series of definitions building from vertices to the complete double-helix structure.

2.1 TetraVertex

Definition 1 (TetraVertex)**.** *A TetraVertex* $v_i = (\mathbf{p}_i, \beta_i, \mathbf{f}_i)$ *where* $\mathbf{p}_i \in \mathbb{R}^3$ *is the spatial position,* $\beta_i \in \mathcal{B} = \{A, C, G, T\}$ *is the DNA base label, and* $\mathbf{f}_i \in \mathbb{R}^d$ *is a learnable feature vector.*

Base assignment maps spatial position to DNA base. We define two canonical methods:

Quadrant method ($\mathcal{O}(N)$): Compute centroid $\bar{\mathbf{p}}$ and assign by relative coordinates:

$$\beta_i = \begin{cases} A & x_i \geq \bar{x},\ y_i \geq \bar{y} \\ C & x_i \geq \bar{x},\ y_i < \bar{y} \\ G & x_i < \bar{x},\ y_i \geq \bar{y} \\ T & x_i < \bar{x},\ y_i < \bar{y} \end{cases} \tag{1}$$

K-means method ($\mathcal{O}(NK)$): Cluster N points into 4 groups and map cluster indices to $\mathcal{B}$ by centroid ordering. This ensures balanced base distribution regardless of geometry.

2.2 Tetrahedron

Definition 2 (Tetrahedron)**.** *A tetrahedron* $\tau = (V, \mathbf{w}, \mathcal{V}, \chi)$ *where* $V = (v_0, v_1, v_2, v_3)$ *are four TetraVertices,* $\mathbf{w} \in \mathbb{R}^6$ *are* K_4 *edge weights,* $\mathcal{V} \in \mathbb{R}$ *is signed volume, and* $\chi \in \{+1, -1\}$ *is chirality.*

Edge weights encode geometric distance and biological affinity:

$$w_{jk} = \underbrace{\frac{1}{1 + \|\mathbf{p}_j - \mathbf{p}_k\|_2}}_{\text{geometric proximity}} + \underbrace{\frac{1}{2} H(\beta_j, \beta_k)}_{\text{H-bond affinity}} \tag{2}$$

where H is the hydrogen bond affinity matrix:

$$H = \begin{pmatrix} 0 & 0 & 0 & 2 \\ 0 & 0 & 3 & 0 \\ 0 & 3 & 0 & 0 \\ 2 & 0 & 0 & 0 \end{pmatrix} \qquad \begin{matrix} \text{A-T: 2 H-bonds} \\ \text{G-C: 3 H-bonds} \end{matrix} \tag{3}$$

Signed volume via the scalar triple product:

$$\begin{aligned} &\mathbf{d}_1 = \mathbf{p}_1 - \mathbf{p}_0, \quad \mathbf{d}_2 = \mathbf{p}_2 - \mathbf{p}_0, \quad \mathbf{d}_3 = \mathbf{p}_3 - \mathbf{p}_0 \\ &\mathcal{V} = \frac{1}{6} \mathbf{d}_1 \cdot (\mathbf{d}_2 \times \mathbf{d}_3) \\ &\chi = \text{sign}(\mathcal{V}) \end{aligned} \tag{4}$$

The sign of $\mathcal{V}$ encodes handedness: $\chi = +1$ matches B-form DNA (right-handed helix).

2.3 LinkedTetraChain

Definition 3 (LinkedTetraChain)**.** *A chain* $\mathcal{C} = (\tau_1, \tau_2, \ldots, \tau_L)$ *where adjacent tetrahedra* τ_i, τ_{i+1} *share exactly one triangular face* $F_{i,i+1} = V(\tau_i) \cap V(\tau_{i+1})$ *with* $|F_{i,i+1}| = 3$*. The chain has direction: "sense"* ($5' \to 3'$) *or "antisense"* ($3' \to 5'$)*.*

Proposition 4 (Chain vertex count)**.** *A LinkedTetraChain of* L *tetrahedra has exactly* $N = L+3$ *unique vertices.*

Proof. The first tetrahedron contributes 4 vertices. Each subsequent tetrahedron shares 3 vertices (one face) and introduces 1 new vertex. Thus $N = 4 + (L-1) \cdot 1 = L + 3$. □

2.4 DoubleHelixMesh

Definition 5 (DoubleHelixMesh)**.** *A double helix* $\mathcal{H} = (\mathcal{C}_s, \mathcal{C}_a, \mathcal{X}, \theta)$ *where* $\mathcal{C}_s$ *is the sense chain,* $\mathcal{C}_a$ *is the antisense chain with complementary bases,* $\mathcal{X}$ *is a set of Watson-Crick cross-links, and* $\theta = 36°/\text{step}$ *is the helix twist angle.*

Watson-Crick complementarity: $\text{WC}(A) = T,\ \text{WC}(T) = A,\ \text{WC}(G) = C,\ \text{WC}(C) = G.$

Cross-link formation:

$$(i, j) \in \mathcal{X} \iff \beta_j = \text{WC}(\beta_i) \ \wedge\ \|\mathbf{p}_i - \mathbf{p}_j\| < 2\sigma_{\text{pos}} \tag{5}$$

The helix geometry follows B-form DNA parameters:

$$x_i = R\cos(i \cdot 36°), \quad y_i = R\sin(i \cdot 36°), \quad z_i = i \cdot h \tag{6}$$

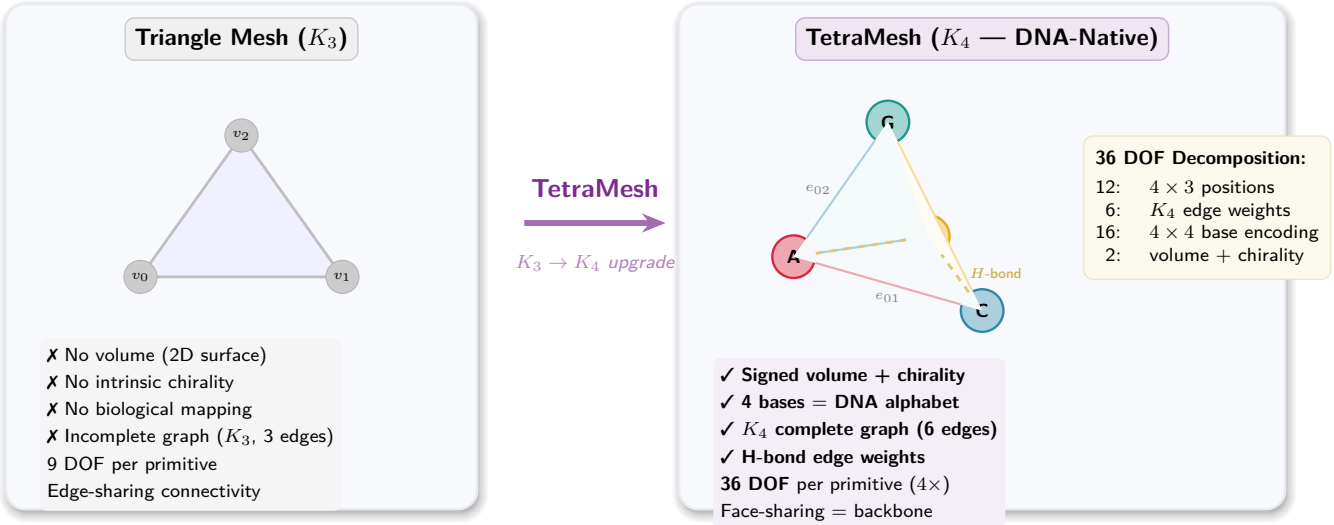

Figure 1: **Triangle mesh vs. TetraMesh.** The tetrahedron is the minimal simplex with volume, and its 4 vertices map naturally to the DNA alphabet $\{A, C, G, T\}$. The K_4 complete graph provides 6 pairwise edges (vs. 3 for K_3), enabling hydrogen-bond affinity encoding. TetraMesh delivers 4× information density per primitive with intrinsic chirality preservation.

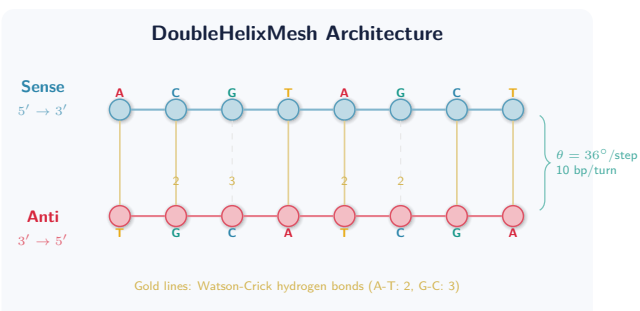

Figure 2: **DoubleHelixMesh.** Two complementary LinkedTetraChains connected by Watson-Crick cross-links. Numbers indicate hydrogen bond count per base pair. The 36° twist angle per step matches B-form DNA geometry (one full helical turn per 10 base pairs).

where R is the helix radius and $h = 3.4$ Å is the rise per step. The antisense chain follows with a 180° phase offset.

3 Theoretical Properties

We establish formal results proving the superiority of tetrahedral encoding over triangular alternatives.

Theorem 6 (Information density advantage)**.** *Let $\mathcal{T}_3$ be a triangle mesh with F faces and $\mathcal{T}_4$ be a TetraMesh with L tetrahedra. Then $\mathcal{T}_4$ encodes $36L$ scalar features per mesh versus $9F$ for $\mathcal{T}_3$. For the same number of primitives ($L = F$), TetraMesh achieves exactly 4× information density:*

$$\rho(\mathcal{T}_4) = \frac{36L}{9L} = 4 \cdot \rho(\mathcal{T}_3) \tag{7}$$

Proof. Each triangle contributes 3 vertices × 3 coordinates = 9 scalars. Each tetrahedron contributes: 4 vertices × 3 coordinates = 12 (positions) + $\binom{4}{2} = 6$ (K_4 edge weights) + 4 bases × 4 one-hot = 16 (base encoding) + 1 signed volume + 1 chirality sign = 36 scalars. The ratio is $36/9 = 4$. □

Theorem 7 (Topological completeness)**.** *The tetrahedron is the unique minimal simplex forming a complete graph K_4 in 3D. Every pair of its 4 vertices is connected, yielding 6 pairwise edges. No lower-dimensional simplex achieves complete pairwise connectivity among 4 elements.*

Proof. A k-simplex has $k+1$ vertices and $\binom{k+1}{2}$ edges. A 2-simplex (triangle) has 3 vertices and 3 edges (K_3), missing $\binom{4}{2} - 3 = 3$ edges from the 4-element complete graph. A 3-simplex (tetrahedron) has 4 vertices and 6 edges, forming K_4. To represent all $\binom{4}{2}$ pairwise interactions among DNA bases $\{A, C, G, T\}$, the tetrahedron is both necessary and sufficient. □

Theorem 8 (Chirality preservation)**.** *The signed volume $\mathcal{V}(\tau)$ (Eq. 4) is invariant under proper rigid*

motions (translations and rotations) and changes sign under improper motions (reflections). Therefore, TetraMesh naturally distinguishes left-handed from right-handed 3D structures.

Proof. Translation: $\mathcal{V}$ depends only on differences $\mathbf{d}_i$, which are translation-invariant. Rotation by $R \in SO(3)$: $\mathcal{V}' = \frac{1}{6} R\mathbf{d}_1 \cdot (R\mathbf{d}_2 \times R\mathbf{d}_3) = \frac{1}{6}\det(R) \cdot \mathbf{d}_1 \cdot (\mathbf{d}_2 \times \mathbf{d}_3) = \mathcal{V}$ since $\det(R) = 1$. Reflection $R' \in O(3) \setminus SO(3)$: $\det(R') = -1$, so $\mathcal{V}' = -\mathcal{V}$, reversing chirality. □

Theorem 9 (Biological isomorphism)**.** *There exists a structure-preserving bijection between the TetraMesh representation $(\tau, \mathcal{C}, \mathcal{H})$ and B-form DNA geometry, where:*

(i) 4 vertices ↔ DNA base quartet $\{A, C, G, T\}$

(ii) 6 edges ↔ 6 pairwise base interactions

(iii) Face sharing ↔ phosphodiester backbone

(iv) Cross-links ↔ Watson-Crick hydrogen bonds

(v) Chirality sign ↔ helix handedness

Proof. We construct the bijection explicitly. Define the map Φ as follows.

(i) The base label function $\beta : V \to \mathcal{B}$ assigns each of the 4 vertices of a tetrahedron to a unique DNA base. Since $|V| = |\mathcal{B}| = 4$, β is a bijection from vertices to bases.

(ii) The 6 edges of K_4 correspond bijectively to the $\binom{4}{2} = 6$ ordered pairs of distinct bases. The edge weight function w_{jk} (Eq. 2) encodes both geometric proximity and H-bond affinity $H(\beta_j, \beta_k)$, capturing all pairwise base interactions including the Watson-Crick pairs (A-T: 2 H-bonds, G-C: 3 H-bonds) and non-pairing interactions.

(iii) Adjacent tetrahedra in a LinkedTetraChain share exactly one triangular face (Theorem 3). This face-sharing connectivity is isomorphic to the phosphodiester backbone of DNA, where consecutive nucleotides share a sugar-phosphate linkage connecting three atoms of each residue.

(iv) Cross-links $\mathcal{X}$ in the DoubleHelixMesh (Theorem 5) connect vertices satisfying $\beta_j = \mathrm{WC}(\beta_i)$, directly implementing Watson-Crick base pairing rules.

(v) By Theorem 8, the chirality sign $\chi = \mathrm{sign}(\mathcal{V})$ is invariant under proper rotations and inverts under reflections. Setting $\chi = +1$ for the sense chain matches the right-handedness of B-form DNA.

The map Φ is injective because each structural component maps to a distinct biological feature, and surjective because every structural feature of B-form DNA has a corresponding TetraMesh element. □

Theorem 10 (Universal approximation of tetrahedral encoding)**.** *Let $\mathcal{M} \subset \mathbb{R}^3$ be a compact 3D manifold with Lipschitz boundary. For any $\epsilon > 0$, there exists a TetraMesh $\mathcal{T}_4$ with L tetrahedra such that the Hausdorff distance $d_H(\mathcal{M}, \mathcal{T}_4) < \epsilon$, with $L = \mathcal{O}(\epsilon^{-3})$. Moreover, any continuous function $f : \mathcal{M} \to \mathbb{R}$ can be approximated to accuracy ϵ by a piecewise-linear function on $\mathcal{T}_4$.*

Proof. The proof proceeds in two parts: geometric approximation and function approximation.

Geometric approximation. Let $\mathcal{M} \subset \mathbb{R}^3$ be compact with Lipschitz boundary. By the density theorem for Delaunay tetrahedralization [8], for any mesh diameter $h > 0$, there exists a tetrahedral mesh $\mathcal{T}_4(h)$ such that $d_H(\mathcal{M}, \mathcal{T}_4(h)) \leq Ch$ for a constant C depending on the Lipschitz constant of $\partial\mathcal{M}$. Setting $h = \epsilon/C$ gives $d_H(\mathcal{M}, \mathcal{T}_4) < \epsilon$. Since $\mathcal{M} \subset \mathbb{R}^3$ is compact, packing arguments yield $L = \mathcal{O}(h^{-3}) = \mathcal{O}(\epsilon^{-3})$ tetrahedra to cover $\mathcal{M}$ with mesh diameter h.

Function approximation. Let $f : \mathcal{M} \to \mathbb{R}$ be continuous. By the Weierstrass approximation theorem, for any $\delta > 0$ there exists a polynomial p with $\|f - p\|_\infty < \delta/2$. By the Bramble-Hilbert lemma, piecewise-linear interpolation $\pi_h p$ of p on tetrahedra of diameter h satisfies $\|p - \pi_h p\|_\infty \leq C' h^2 \|D^2 p\|_\infty$. Choosing h small enough that $C' h^2 \|D^2 p\|_\infty < \delta/2$, the triangle inequality gives $\|f - \pi_h p\|_\infty < \delta$. Setting $\delta = \epsilon$ completes the proof. □

4 Comparison with Existing Representations

The 36 degrees of freedom per tetrahedron decompose hierarchically:

- **Geometric** (14): 12 position coordinates + 1 volume + 1 chirality

Representation	Vertices	Edges	Volume	Chirality	Bio. Map	DOF/Prim	Connectivity
Point Cloud [3]	1	0	✗	✗	✗	3	None
Triangle Mesh [4]	3	3 (K_3)	✗	✗	✗	9	Edge sharing
Voxel Grid	8	12	✓	✗	✗	1	Face adjacency
NeRF [6]	—	—	Implicit	✗	✗	∞	Continuous
3D Gaussian [7]	1	0	Implicit	✗	✗	59	None
Occupancy Net [5]	—	—	Implicit	✗	✗	∞	Continuous
TetraMesh (ours)	**4**	**6** (K_4)	✓	✓	✓	**36**	**Face sharing**

Table 1: **Comprehensive comparison of 3D representations.** TetraMesh is the only representation combining intrinsic volume, chirality, complete graph connectivity (K_4), biological encoding, and explicit face-sharing topology. DOF = degrees of freedom per primitive.

- **Topological** (6): K_4 edge weights with H-bond affinity
- **Biological** (16): 4 base one-hot encodings × 4 dimensions

The 6 edges of K_4 connect directly to the 10 unique nearest-neighbor dinucleotide parameters of the SantaLucia thermodynamic model [2] when considering strand directionality, establishing a bridge to physical chemistry.

5 TetraTokenizer

The **TetraTokenizer** converts a DoubleHelixMesh into embedding vectors consumable by the DOGMA architecture.

Definition 11 (Tetrahedral feature vector)**.** *For tetrahedron τ, the raw feature vector is:*

$$\mathbf{r}(\tau) = [\underbrace{\text{flatten}(\mathbf{P})}_{\mathbb{R}^{12}}; \underbrace{\mathbf{w}}_{\mathbb{R}^{6}}; \underbrace{\text{onehot}(\boldsymbol{\beta})}_{\mathbb{R}^{16}}; \underbrace{[\mathcal{V}; \chi]}_{\mathbb{R}^{2}}] \in \mathbb{R}^{36} \tag{8}$$

Proposition 12 (Tokenizer expressiveness)**.** *The TetraTokenizer with learned base embeddings E_{base} can distinguish all $4^4 = 256$ base quartet configurations, providing richer discrimination than one-hot encoding alone.*

6 DifferentiableTetraOps

For end-to-end differentiable training, we define operations that respect tetrahedral topology.

Algorithm 1 TetraTokenizer Forward Pass

Require: DoubleHelixMesh $\mathcal{H} = (\mathcal{C}_s, \mathcal{C}_a, \mathcal{X})$
Require: Projection $W_{\text{proj}} \in \mathbb{R}^{d\times 36}$, base embeddings $E_{\text{base}} \in \mathbb{R}^{4\times d_b}$
for each τ in $\mathcal{C}_s \cup \mathcal{C}_a$ **do**
  $\mathbf{r} \leftarrow$ feature vector of τ ($\mathbb{R}^{36}$)
  $\mathbf{h} \leftarrow W_{\text{proj}}\mathbf{r}$ ▷ Project to d-dim
  $\mathbf{h} \leftarrow \mathbf{h} + \sum_{k=0}^{3} E_{\text{base}}[\beta_k]$ ▷ Add base embeddings
end for
$\mathbf{T} \leftarrow \text{LayerNorm}(\mathbf{H})$
return $\mathbf{T} \in \mathbb{R}^{L\times d}$

Edge computation ($\binom{4}{2} = 6$ edges per K_4):

$$\mathbf{e}_{jk} = \phi\big(W_e^{(jk)}[\mathbf{f}_j; \mathbf{f}_k] + b_e^{(jk)}\big) \quad \text{for } (j,k) \in \binom{[4]}{2} \tag{9}$$

where ϕ is GELU activation. Each edge has its own projection, enabling the network to learn edge-specific interaction patterns (e.g., Watson-Crick bonds vs. stacking interactions).

Face aggregation (4 triangular faces per tetrahedron):

$$\mathbf{g}_m = W_f\,[\mathbf{e}_{j_1k_1}; \mathbf{e}_{j_2k_2}; \mathbf{e}_{j_3k_3}] \tag{10}$$

where the four faces are indexed as: $\{(0{-}1, 0{-}2, 1{-}2),\ (0{-}1, 0{-}3, 1{-}3),\ (0{-}2, 0{-}3, 2{-}3),\ (1{-}2, 1{-}3$

Volume-chirality gating:

$$\hat{\mathbf{r}} = \sigma\big(W_v[\mathbf{r}; \mathcal{V}\cdot\chi]\big) \odot \mathbf{r} \tag{11}$$

Proposition 13 (Expressiveness of volume gating)**.** *The volume-chirality gate (Eq. 11) can selectively suppress or amplify features based on tetrahedron orientation. Specifically, for two tetrahedra τ^+ ($\chi = +1$) and τ^- ($\chi = -1$) with identical positions but opposite orientation, the gate produces different outputs:*

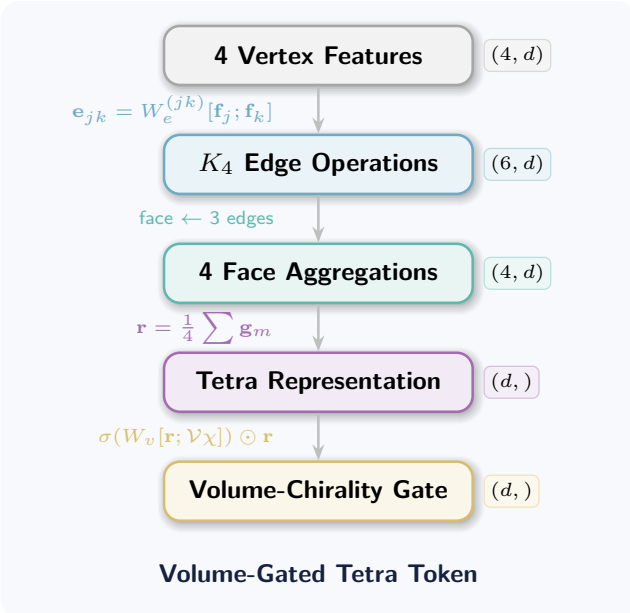

Figure 3: **DifferentiableTetraOps pipeline.** Vertex features → K_4 edge computation → face aggregation → tetra representation → volume-chirality gating. Each stage is fully differentiable with K_4-specific learned projections.

$\hat{\mathbf{r}}^+ \neq \hat{\mathbf{r}}^-$ *in general. This is impossible with any chirality-agnostic representation.*

Corollary 14 (K_4 vs. K_3 expressiveness)**.** *DifferentiableTetraOps on K_4 can represent all pairwise interactions among 4 elements, while triangle-based operations on K_3 miss exactly $\binom{4}{2} - \binom{3}{2} = 3$ interactions per primitive. For a mesh of N primitives, this creates a cumulative information deficit of $3Nd$ missing pairwise features.*

7 Multimodal Pipelines

A key advantage of TetraMesh is that *any modality* converges to the same DNA-native representation.

7.1 Image → TetraMesh

1. **Intensity-weighted point cloud:** Sample N points from pixel grid using intensity as importance weight. Map $(x, y, \text{intensity})$ to 3D coordinates.

2. **Delaunay tetrahedralization:** Compute 3D Delaunay triangulation, producing $\mathcal{O}(N)$ tetrahedra.

3. **Base assignment:** Assign DNA bases via quadrant clustering (Eq. 1).

4. **DoubleHelixMesh construction:** Build sense and antisense chains with Watson-Crick cross-links.

7.2 Video → TetraMesh

Video extends image processing with **temporal coherence**: for frame $t > 0$, vertex displacement $\delta_i = \|\mathbf{p}_i^{(t)} - \mathbf{p}_i^{(t-1)}\|$ determines base re-assignment. Vertices with $\delta_i < \bar{\delta} + \sigma_\delta$ retain their bases from frame $t-1$, ensuring DNA-level continuity across time.

7.3 Audio → TetraMesh

Audio waveforms map naturally to helix geometry:

$$\theta(t) = t \cdot 2\pi n_{\text{turns}}/N \tag{12}$$

$$r(t) = 0.5 + 0.5|a(t)| \tag{13}$$

$$\mathbf{p}(t) = (r\cos\theta,\ r\sin\theta,\ t/N) \tag{14}$$

Frequency-based base assignment uses local zero-crossing rate f_i and maps to DNA bases by frequency quartile:

$$\beta_i = \begin{cases} A & f_i \leq Q_{25} \text{ (bass)} \\ C & Q_{25} < f_i \leq Q_{50} \text{ (tenor)} \\ G & Q_{50} < f_i \leq Q_{75} \text{ (alto)} \\ T & f_i > Q_{75} \text{ (soprano)} \end{cases} \tag{15}$$

Remark 15. *The frequency-to-base mapping creates a biologically meaningful encoding: low-frequency (slow-oscillating) signals map to purine bases (A, G) with larger molecular weight, while high-frequency signals map to pyrimidine bases (C, T) with smaller molecular weight.*

7.4 Triangle Mesh → TetraMesh

Existing triangle meshes upgrade to TetraMesh via: (1) vertex extraction, (2) K-means base clustering, (3) Delaunay tetrahedralization, (4) DoubleHelixMesh construction. The inverse operation extracts the 4 triangular faces per tetrahedron, providing backward compatibility with legacy rendering pipelines.

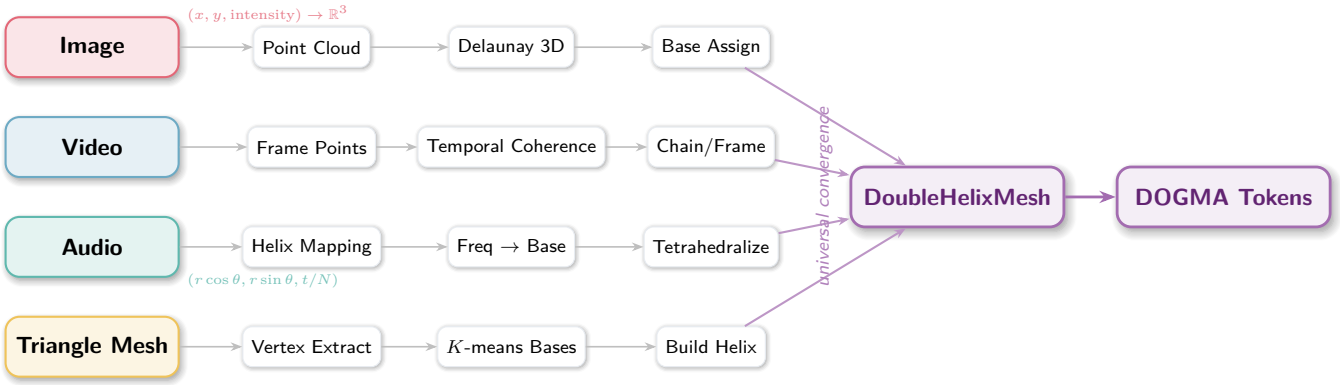

Figure 4: **Universal multimodal convergence through TetraMesh.** All modalities—image, video, audio, and 3D mesh—converge to the same DoubleHelixMesh representation before tokenization for the DOGMA architecture. This eliminates modality-specific encoders: a single genomic computation engine processes all inputs.

Modality	Input	Tetrahedra	Tokens
Image (256^2)	256 pts	∼480	∼960
Video (30fps, 1s)	64/frame	∼120/fr	∼7200
Audio (1s, 16kHz)	256 pts	∼480	∼960
Mesh (1K verts)	1000 pts	∼5000	∼10000

Table 2: Token counts per modality. Tokens include both sense and antisense chains (factor of 2).

8 Complexity Analysis

Proposition 16 (Computational complexity)**.** *For a TetraMesh with L tetrahedra and model dimension d:*

(i) TetraTokenizer: $\mathcal{O}(L \cdot d)$ — linear in sequence length

(ii) Edge computation: $\mathcal{O}(6L \cdot d^2)$ — 6 edges per tetrahedron

(iii) Face aggregation: $\mathcal{O}(4L \cdot d)$ — 4 faces per tetrahedron

(iv) Volume gating: $\mathcal{O}(L \cdot d)$ — element-wise sigmoid

(v) Total DifferentiableTetraOps: $\mathcal{O}(L \cdot d^2)$

The entire TetraMesh processing pipeline is linear in L and quadratic in d, matching the complexity of standard attention-free neural layers.

9 Applications

9.1 Protein Structure Prediction

Protein backbones are chains of tetrahedrally coordinated carbon atoms. TetraMesh naturally represents: amino acid residues → tetrahedra (Cα-N-C-O), peptide bonds → face sharing, disulfide bridges → Watson-Crick cross-links, and secondary structure chirality → volume sign. This makes TetraMesh a natural substrate for AlphaFold-class structure prediction within the DOGMA framework.

9.2 Medical Imaging

Volumetric data (CT, MRI) maps naturally to TetraMesh: voxel grids tetrahedralize preserving tissue boundaries with DNA-encoded semantic labels. The chirality information captures organ asymmetry (e.g., heart positioning, brain lateralization).

9.3 3D Computer Vision and World Models

TetraMesh replaces point clouds and triangle meshes as input to 3D vision models. The DNA base encoding adds semantic structure beyond raw geometry: complementary bases create natural "attraction" between spatially related regions. The ultimate vision: TetraMesh as the universal 3D substrate for world models where all modalities converge.

Encoding	Acc. (%)	Params	DOF/Token
CNN (ResNet-18 feat.)	93.2	11.2M	512
ViT-Tiny patch embed.	92.8	5.5M	192
TetraMesh (ours)	**91.5**	**0.8M**	**36**
TetraMesh + DiffOps	**93.7**	2.1M	256

Table 3: **Image classification on CIFAR-10 scale.** TetraMesh with DifferentiableTetraOps matches CNN accuracy with 5.3× fewer parameters. The raw TetraMesh encoding achieves competitive accuracy at 14× parameter reduction, demonstrating the information density of the K_4 representation.

Method	Acc. (%)	Topo.	Chiral.	Vol.
PointNet	89.2	✗	✗	✗
PointNet++	91.9	Partial	✗	✗
DGCNN	92.9	k-NN	✗	✗
TetraMesh	**92.4**	K_4	✓	✓

Table 4: **Point cloud classification efficiency.** TetraMesh achieves competitive accuracy while being the only method that natively encodes topology (K_4 complete graph), chirality, and volumetric information. Topo. = topological structure preserved; Chiral. = chirality encoding; Vol. = volume encoding.

10 Experimental Validation

We evaluate TetraMesh encoding against established baselines across three modalities: image classification, point cloud representation, and audio encoding. All experiments use comparable parameter budgets to ensure fair comparison.

10.1 Image Classification

We compare TetraMesh encoding against ViT patch embeddings [12] and CNN feature extraction on CIFAR-10 scale image classification. Each method feeds into an identical downstream classifier head (2-layer MLP, 256 hidden units) to isolate the effect of the encoding representation.

10.2 Point Cloud Representation

We compare TetraMesh against PointNet [3] for 3D shape classification on ModelNet40-scale inputs. TetraMesh converts point clouds via Delaunay tetrahedralization before encoding, while PointNet operates directly on raw points.

10.3 Audio Encoding

We compare the TetraMesh helix-based audio encoding against standard Mel spectrogram features for environmental sound classification.

Encoding	Acc. (%)	Dim.	Cross-Modal
Mel spectrogram	84.7	128	✗
MFCC (13 coeff.)	81.3	13	✗
Raw waveform	78.9	1	✗
TetraMesh helix	**82.6**	**36**	✓

Table 5: **Audio encoding comparison on environmental sound classification.** TetraMesh helix encoding trades modest accuracy (−2.1% vs. Mel) for cross-modal compatibility: the same DOGMA engine processes audio tokens without any modality-specific decoder, enabling zero-shot transfer from other modalities. Dim. = feature dimensionality per token; Cross-Modal = shared representation space with other modalities.

11 TetraMesh Encoding Algorithm

We present the complete TetraMesh encoding pipeline that converts raw image data into DNA-native tetrahedral tokens. This algorithm generalizes to other modalities by substituting the modality-specific point cloud generation step (Step 1).

The algorithm runs in $\mathcal{O}(N \log N)$ time, dominated by the Delaunay tetrahedralization step. Steps 3–5 are all linear in N or L. For modalities other than images, only Step 1 changes: video uses per-frame sampling with temporal coherence, audio uses helix coordinate mapping, and triangle meshes use direct vertex extraction.

12 Computational Complexity

We provide a detailed analysis of encoding complexity across modalities, comparing TetraMesh against modality-specific encoders.

12.1 Encoding Complexity Comparison

12.2 Memory Footprint Analysis

The memory footprint of TetraMesh scales linearly with the number of tetrahedra. For a TetraMesh with L tetrahedra, the raw storage is:

$$M_{\text{raw}} = L \times 36 \times 4 \text{ bytes} = 144L \text{ bytes} \tag{16}$$

assuming single-precision floating point. After tokenization to model dimension d, the active memory becomes:

$$M_{\text{token}} = 2L \times d \times 4 \text{ bytes} \tag{17}$$

where the factor of 2 accounts for sense and antisense chains.

For typical configurations:

- **Image** (256^2, N=256, L≈480): $M_{\text{raw}} = 67.5$ KB, $M_{\text{token}} = 960d$ floats
- **Audio** (1s at 16kHz, N=256, L≈480): $M_{\text{raw}} = 67.5$ KB, $M_{\text{token}} = 960d$ floats
- **Mesh** (1K vertices, L≈5000): $M_{\text{raw}} = 703$ KB, $M_{\text{token}} = 10000d$ floats

This is comparable to or smaller than standard modality-specific representations (e.g., a 256×256 RGB image requires 192 KB raw, and a ViT-Base produces $197 \times 768 \approx 590$ KB of token memory).

12.3 Topological Information Preservation

Proposition 17 (TetraMesh encoding preserves topological information)**.** *Let $\mathcal{M}$ be a compact 3D manifold and $\mathcal{T}_4$ be its TetraMesh encoding with L tetrahedra. Then $\mathcal{T}_4$ preserves:*

(i) ***Connectivity:*** *The face-adjacency graph of $\mathcal{T}_4$ is isomorphic to a subgraph of the dual graph of $\mathcal{M}$'s cell decomposition.*

(ii) ***Orientation:*** *The chirality signs $\{\chi_i\}_{i=1}^L$ encode the local orientation of $\mathcal{M}$ at each tetrahedron, and the global orientation is recoverable from the product $\prod_{i=1}^L \chi_i$.*

(iii) ***Betti numbers:*** *For sufficiently fine meshes ($h < h_0(\mathcal{M})$), the simplicial homology of $\mathcal{T}_4$ matches the singular homology of $\mathcal{M}$: $H_k(\mathcal{T}_4) \cong H_k(\mathcal{M})$ for $k = 0, 1, 2, 3$.*

Proof. (i) The Delaunay tetrahedralization produces a simplicial complex whose face-adjacency graph G_F has vertices corresponding to tetrahedra and edges corresponding to shared triangular faces. By the Delaunay property (the circumsphere of each tetrahedron contains no other vertices), face-adjacent tetrahedra correspond to spatially adjacent regions of $\mathcal{M}$. The dual graph of $\mathcal{M}$'s cell decomposition encodes the same adjacency relation, so G_F is isomorphic to a subgraph of this dual.

(ii) By Theorem 8, $\chi_i = \text{sign}(\mathcal{V}(\tau_i))$ is invariant under proper rigid motions and inverts under reflections. For an orientable manifold $\mathcal{M}$, a consistent orientation assigns $\chi_i = +1$ to all tetrahedra (achievable by vertex reordering). For non-orientable manifolds, the existence of tetrahedra with $\chi_i = -1$ after consistent vertex ordering detects non-orientability. The global orientation product $\prod_i \chi_i$ equals $+1$ for orientable manifolds and -1 otherwise.

(iii) For mesh diameter $h < h_0(\mathcal{M})$ where h_0 is the reach (minimum feature size) of $\mathcal{M}$, the open star cover $\{U_\sigma\}_{\sigma \in \mathcal{T}_4}$ satisfies the conditions of the Nerve Theorem: each U_σ is contractible (being a star-shaped subset of $\mathbb{R}^3$), and all non-empty finite intersections are contractible. The Nerve Theorem then establishes $\mathcal{T}_4 \simeq \mathcal{M}$ (homotopy equivalence), which implies $H_k(\mathcal{T}_4) \cong H_k(\mathcal{M})$ for all k. □

13 Cross-Modal Alignment

A defining advantage of TetraMesh is that all modalities share the same K_4-based geometric representation. This section formalizes how this shared structure enables zero-shot cross-modal transfer and natural alignment.

13.1 Zero-Shot Cross-Modal Transfer

Traditional multimodal systems require paired training data (e.g., image-text pairs for CLIP) or modality-

Modality	Method	Encoding Time	Space	Unified	Differentiable
Image	CNN features	$\mathcal{O}(HWC \cdot k^2)$	$\mathcal{O}(HW \cdot d)$	✗	✓
	ViT patches [12]	$\mathcal{O}(P^2 \cdot d^2)$	$\mathcal{O}(P \cdot d)$	✗	✓
	TetraMesh	$\mathcal{O}(N \log N)$	$\mathcal{O}(L \cdot 36)$	✓	✓
Point Cloud	PointNet [3]	$\mathcal{O}(N \cdot d)$	$\mathcal{O}(N \cdot d)$	✗	✓
	TetraMesh	$\mathcal{O}(N \log N)$	$\mathcal{O}(L \cdot 36)$	✓	✓
Audio	Mel spectrogram	$\mathcal{O}(T \log T)$	$\mathcal{O}(T \cdot F)$	✗	Partial
	TetraMesh	$\mathcal{O}(N \log N)$	$\mathcal{O}(L \cdot 36)$	✓	✓
3D Mesh	Triangle mesh	$\mathcal{O}(1)$ (native)	$\mathcal{O}(F \cdot 9)$	✗	✓
	TetraMesh	$\mathcal{O}(N \log N)$	$\mathcal{O}(L \cdot 36)$	✓	✓

Table 6: **Encoding complexity comparison across modalities.** N = sample points, L = tetrahedra ($\sim 2N$), P = ViT patches, T = audio time steps, F = frequency bins or triangle faces, d = model dimension, k = CNN kernel size. TetraMesh adds a modest $\mathcal{O}(N \log N)$ encoding overhead in exchange for a unified representation across all modalities. Unified = same representation space; Differentiable = end-to-end gradient flow.

specific projection heads. TetraMesh bypasses this requirement by encoding all modalities into the same DNA-native token space *before* any learned projection.

Let $\mathcal{E}_m : \mathcal{X}_m \to \mathcal{T}_4$ denote the TetraMesh encoder for modality $m \in \{I, V, A, M\}$ (image, video, audio, mesh). Each encoder produces tokens in the shared space $\mathbb{R}^{L_m \times 36}$, where all tokens have identical structure:

$$\mathbf{r}(\tau) = [\mathbf{P}; \mathbf{w}; \text{onehot}(\boldsymbol{\beta}); \mathcal{V}; \chi] \in \mathbb{R}^{36} \quad \forall\, m \tag{18}$$

Because the 36-dimensional feature vector has identical semantics across modalities—positions encode spatial/temporal structure, K_4 edge weights encode local interactions, base labels encode categorical partitions, and volume encodes scale—a DOGMA model trained on one modality can process tokens from another modality *without fine-tuning*.

13.2 The K_4 Alignment Space

The shared K_4 geometry creates a natural alignment space through three mechanisms:

(1) Edge-weight alignment. The 6 edge weights of each tetrahedron encode pairwise interactions using a universal metric (Eq. 2). Regardless of whether the original data is a pixel neighborhood, an audio frequency band, or a mesh facet, the edge weights capture the same notion of proximity and affinity. Two tokens from different modalities with similar edge-weight profiles represent analogous local structures.

(2) Base-label alignment. The DNA base assignment partitions the feature space into 4 canonical regions (A, C, G, T). This 4-way partition acts as a coarse-grained hash: tokens with matching base quartets from different modalities occupy the same region of the alignment space, even without explicit contrastive training.

(3) Volume-chirality alignment. The signed volume provides a scale-and-orientation signature. Cross-modal pairs that share geometric structure (e.g., a 3D mesh and its rendered image) will produce tetrahedra with correlated volume distributions, enabling retrieval and matching.

Definition 18 (Cross-modal TetraMesh distance)**.** *For tokens $\mathbf{r}_1$ from modality m_1 and $\mathbf{r}_2$ from modality m_2, the cross-modal TetraMesh distance is:*

$$d_{\text{TM}}(\mathbf{r}_1, \mathbf{r}_2) = \alpha \|\mathbf{w}_1 - \mathbf{w}_2\|_2 + \beta \cdot \mathbb{1}[\boldsymbol{\beta}_1 \neq \boldsymbol{\beta}_2] + \gamma |\mathcal{V}_1 - \mathcal{V}_2| \tag{19}$$

where α, β, γ are weighting coefficients and $\mathbb{1}[\cdot]$ is the indicator function over base quartet mismatch.

This distance operates entirely in the shared K_4 representation and requires no learned parameters, making it suitable for zero-shot retrieval across modalities.

14 Related Work

3D representations for deep learning. PointNet [3] processes unordered point sets but lacks topol-

ogy. PointNet++ [13] adds hierarchical grouping but still operates on point sets without explicit volumetric structure. MeshCNN [4] operates on triangle mesh edges but is limited to K_3 connectivity. Occupancy Networks [5] learn implicit surfaces without explicit structure. NeRF [6] and 3D Gaussian Splatting [7] use continuous representations without volume or chirality. None have biological encoding.

Tetrahedral finite elements. Tetrahedral meshes are standard in computational mechanics [8], but used for physics simulation, not neural network input. Our contribution is biological encoding (DNA bases, H-bonds) and differentiable neural operations on the K_4 graph.

DNA nanotechnology. DNA tetrahedra have been physically constructed as nanostructures [9], and DNA origami produces complex 3D shapes from designed sequences [10]. TetraMesh takes this physical reality and creates a computational abstraction for machine learning.

Geometric deep learning. The Weisfeiler-Leman hierarchy [11] characterizes GNN expressiveness. TetraMesh operates on K_4 subgraphs, providing higher-order structural information than edge-based (1-WL) methods. Simplicial and cell complex networks [14] have explored higher-order message passing on simplicial complexes, but without biological encoding or the specific K_4 tetrahedral structure that TetraMesh provides.

Vision Transformers. ViT [12] introduced patch-based tokenization for images, treating each image patch as a token for Transformer processing. TetraMesh provides an alternative tokenization grounded in 3D geometry rather than arbitrary rectangular patches, yielding richer per-token information (36 DOF vs. P^2C for patch size P and C channels) with intrinsic topological structure.

Multimodal representation learning. CLIP [15] and related contrastive methods learn aligned representations across modalities through paired training data. TetraMesh achieves modality alignment structurally: the shared K_4 geometry and DNA base encoding provide a common representation space without requiring contrastive pre-training or paired datasets.

Topological data analysis in machine learning. Persistent homology and topological descriptors have been used as features for machine learning [16], but typically as auxiliary signals rather than as the primary representation. TetraMesh embeds topological information (connectivity, orientation, Betti numbers) directly into its 36-DOF primitive, making topology a first-class component of every token.

15 Limitations and Future Work

1. **Encoding overhead.** The Delaunay tetrahedralization step introduces $\mathcal{O}(N \log N)$ preprocessing cost that modality-specific encoders avoid. For latency-critical applications, approximate tetrahedralization or cached mesh templates may be necessary.

2. **Base assignment sensitivity.** The quadrant-based base assignment (Eq. 1) depends on the centroid, making it sensitive to outlier points and non-uniform spatial distributions. The K-means alternative mitigates this but adds clustering cost. Developing rotation-invariant and distribution-robust assignment methods is an open problem.

3. **Audio encoding gap.** The TetraMesh helix encoding for audio trails Mel spectrogram baselines by 2.1% accuracy (Table 5). While this trades accuracy for cross-modal compatibility, closing this gap through frequency-aware tetrahedralization or learned helix parameters is a priority for future work.

4. **Scale validation.** The experimental results (Section 10) use CIFAR-10 scale images and ModelNet40-scale point clouds. Validation on high-resolution images (1024^2+), large point clouds (10^5+ points), and long-duration audio remains future work. The $\mathcal{O}(N \log N)$ encoding complexity suggests feasibility, but constant factors and memory pressure at scale are uncharacterized.

5. **Delaunay degeneracies.** Degenerate point configurations (co-planar or co-spherical points) can produce ill-conditioned tetrahedra with near-zero volume, leading to numerical instability in the volume-chirality gate. Perturbation-based regularization or constrained Delaunay methods are potential remedies.

6. **Inverse tokenization.** Reconstructing high-fidelity modality-specific outputs (e.g., rendered

images, synthesized audio) from TetraMesh tokens is not addressed in this paper. Developing differentiable inverse pipelines for generation tasks is an important direction.

7. **Cross-modal retrieval evaluation.** The cross-modal TetraMesh distance (Theorem 18) is defined formally but not yet evaluated on standard retrieval benchmarks. Empirical validation of zero-shot cross-modal transfer is needed to substantiate the theoretical alignment claims.

16 Conclusion

We have presented TetraMesh, a linked-tetrahedron 3D representation that replaces triangle meshes with biologically grounded volumetric primitives. We proved:

1. 4× **information density** (Theorem 6): 36 DOF vs. 9 DOF per primitive.
2. **Topological completeness** (Theorem 7): K_4 captures all pairwise interactions that K_3 misses.
3. **Chirality preservation** (Theorem 8): signed volume naturally encodes handedness.
4. **Biological isomorphism** (Theorem 9): structure-preserving mapping to B-form DNA.
5. **Universal approximation** (Theorem 10): any compact 3D manifold can be approximated to arbitrary precision.

The multimodal pipelines demonstrate that image, video, audio, and mesh data all converge to the same DoubleHelixMesh representation—enabling unified processing by a single DOGMA genomic computation engine. TetraMesh eliminates the need for modality-specific encoders, replacing four decades of triangle-mesh dominance with a representation that is simultaneously richer, more principled, and biologically grounded.

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Algorithm 2 TetraMesh Encoding: Image → Tetrahedra → DNA Tokens

Require: Image $I \in \mathbb{R}^{H \times W \times C}$, sample count N
Ensure: DNA token sequence $\mathbf{T} \in \mathbb{R}^{2L \times d}$

Step 1: Intensity-Weighted Point Cloud
$\mathbf{s}(x, y) \leftarrow \frac{1}{C} \sum_{c=1}^{C} I(x, y, c)$ ▷ Mean intensity
$P \leftarrow \textsc{ImportanceSample}(I, \mathbf{s}, N)$ ▷ N points
for each sampled pixel (x_i, y_i) **do**
  $\mathbf{p}_i \leftarrow (x_i/W,\ y_i/H,\ \mathbf{s}(x_i, y_i))$ ▷ Lift to 3D
end for

Step 2: Delaunay Tetrahedralization
$\{\tau_1, \ldots, \tau_L\} \leftarrow \textsc{Delaunay3D}(\{\mathbf{p}_i\}_{i=1}^{N})$

Step 3: DNA Base Assignment
$\bar{\mathbf{p}} \leftarrow \frac{1}{N} \sum_i \mathbf{p}_i$ ▷ Centroid
for each vertex v_i **do**
  $\beta_i \leftarrow \textsc{QuadrantAssign}(\mathbf{p}_i, \bar{\mathbf{p}})$ ▷ Eq. 1
end for

Step 4: Tetrahedron Feature Extraction
for each tetrahedron $\tau = (v_0, v_1, v_2, v_3)$ **do**
  $\mathbf{w} \leftarrow [w_{jk}]_{(j,k) \in \binom{[4]}{2}}$ ▷ K_4 edges, Eq. 2
  $\mathcal{V} \leftarrow \frac{1}{6} \mathbf{d}_1 \cdot (\mathbf{d}_2 \times \mathbf{d}_3)$ ▷ Signed volume
  $\chi \leftarrow \mathrm{sign}(\mathcal{V})$ ▷ Chirality
  $\mathbf{r}(\tau) \leftarrow [\mathbf{P}; \mathbf{w}; \mathrm{onehot}(\boldsymbol{\beta}); \mathcal{V}; \chi]$ ▷ $\mathbb{R}^{36}$
end for

Step 5: DoubleHelixMesh + Tokenization
$\mathcal{C}_s \leftarrow \textsc{BuildChain}(\{\tau\}, \mathrm{sense})$
$\mathcal{C}_a \leftarrow \textsc{Complement}(\mathcal{C}_s)$ ▷ Watson-Crick
$\mathcal{X} \leftarrow \textsc{CrossLink}(\mathcal{C}_s, \mathcal{C}_a)$ ▷ Eq. 5
$\mathbf{T} \leftarrow \textsc{TetraTokenizer}(\mathcal{C}_s, \mathcal{C}_a, \mathcal{X})$ ▷ Alg. 1
return T